SERKAYON

Digital Diameter Inspection & Capability Analysis Platform

Type: Custom Dashboard with report generation

Client: Hyundai WIA, Chennai

OEM buyer: SP systems, South Korea

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1. Case Study / Initial Situation

The customer was inspecting precision components but relied on manual logs and basic HMI readouts. This caused:

- No centralized history of measurements
- No automated Cp/Cpk capability study
- No traceability across batches, models, and operators
- No quick way to filter OK/NG parts or generate PDF inspection reports
- Frequent delay in detecting drift or out-of-tolerance issues

Bottom line: Quality issues were found late, and audits were painful.

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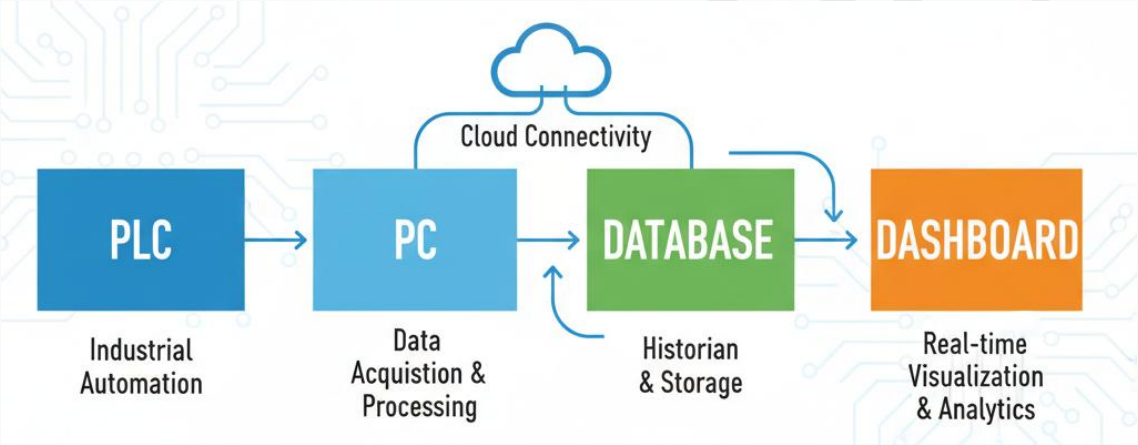
2. System Architecture



A. Data Source

- Accretech Pulcom Diameter inspection unit, Mitsubishi PLC
- **Protocol:** Modbus TCP, MT connect
- **Parameters:** Diameter 1, Diameter 2, Runouts, Part status, QR code

B. Edge Device / Gateway

- Industrial PC running Python polling engine
- Polling interval: 1 second
- Local buffering + retry to prevent data loss

C. Backend Platform

- **PostgreSQL** for historian storage
- **REST API** (FastAPI/Flask) for dashboard and PDF generation
- Automated calculation logic for mean, sigma, Cp, Cpk
- Batch-level analytics engine

D. Frontend / Dashboard

Built with React + Vite + Tailwind:

- Live inspection dashboard
- Historian trend graphs
- OK/NG part counts
- Live model/limit updates
- Cp/Cpk analysis tool
- Model management interface
- PDF reports in OEM-grade format

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3. Key Features Delivered

- **Real-time measurement visualization (1–2 sec)**
- **Automated Cp/Cpk calculator**
  - Calculate by measurement ID
  - Calculate by timestamp range
  - Runout include/exclude toggle
- **Professional PDF export**
  - Trend graph + capability summary
  - OEM-branded layout
- **Interactive results view**
  - Advanced filters (model, date, Lot, status)
  - OK/NG color coding
  - CSV/PDF download
- **Model Management module**
  - Edit limits
  - Add/delete models
- **Custom dashboard per machine**
  - OK/NG indicator
  - Running model
  - Measurement history
- **Batch & supervisor report generation**

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4. Results / Impact

- Inspection reporting time dropped from **30+ minutes → under 10 seconds**
- Drift and tolerance failures detected **immediately**, not hours later
- Audit preparation time reduced by **over 70%**
- Consistent PDF format helped with customer audits and documentation
- Eliminated spreadsheet work and manual math for capability studies

5. Project Duration & Scale

Development Time	Machine Integration	Data Volume
6	**2**	**~2500**
weeks	inspection units	records/hour

6. Screenshots (Summary)

a. Dashboard View

- Historian graph,
- Count of OK/NG parts
- Running model details
- Limit details
- Live measurement update
- OK/NG indicator

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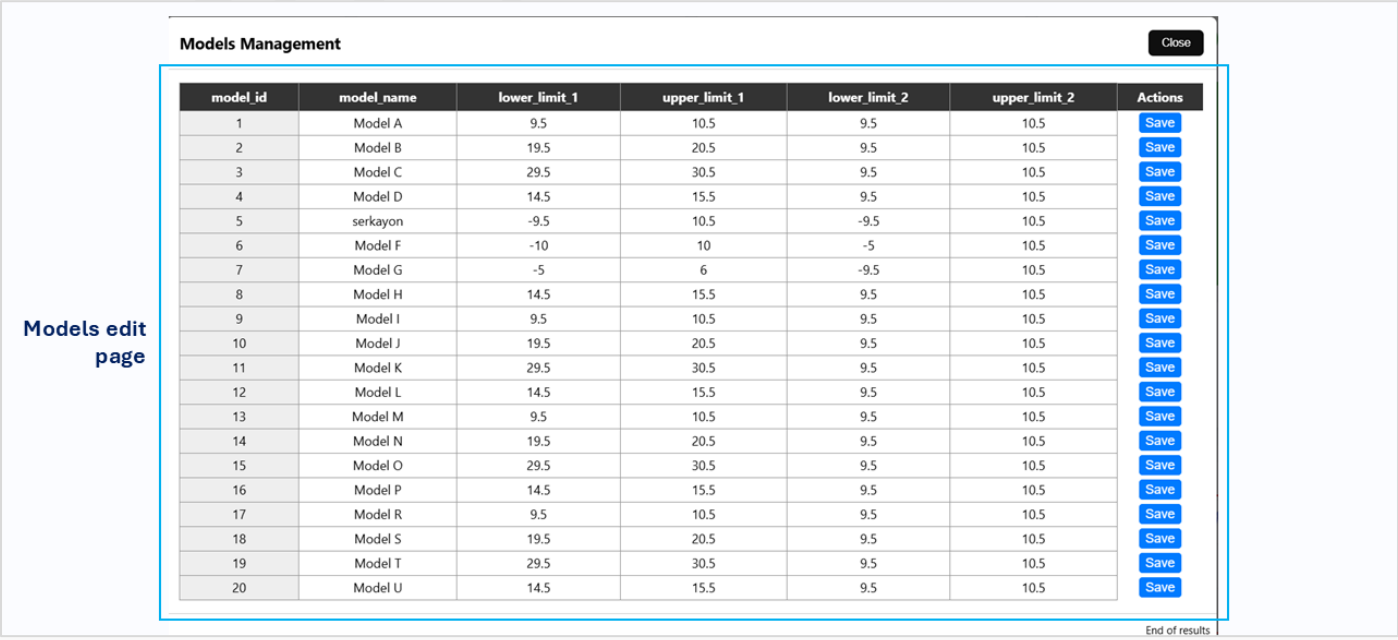
a. Cp / Cpk Calculator

- Inbuilt Cp/Cpk calculator
- Calculate by measure id
- Calculate by time stamp
- Runout disable option

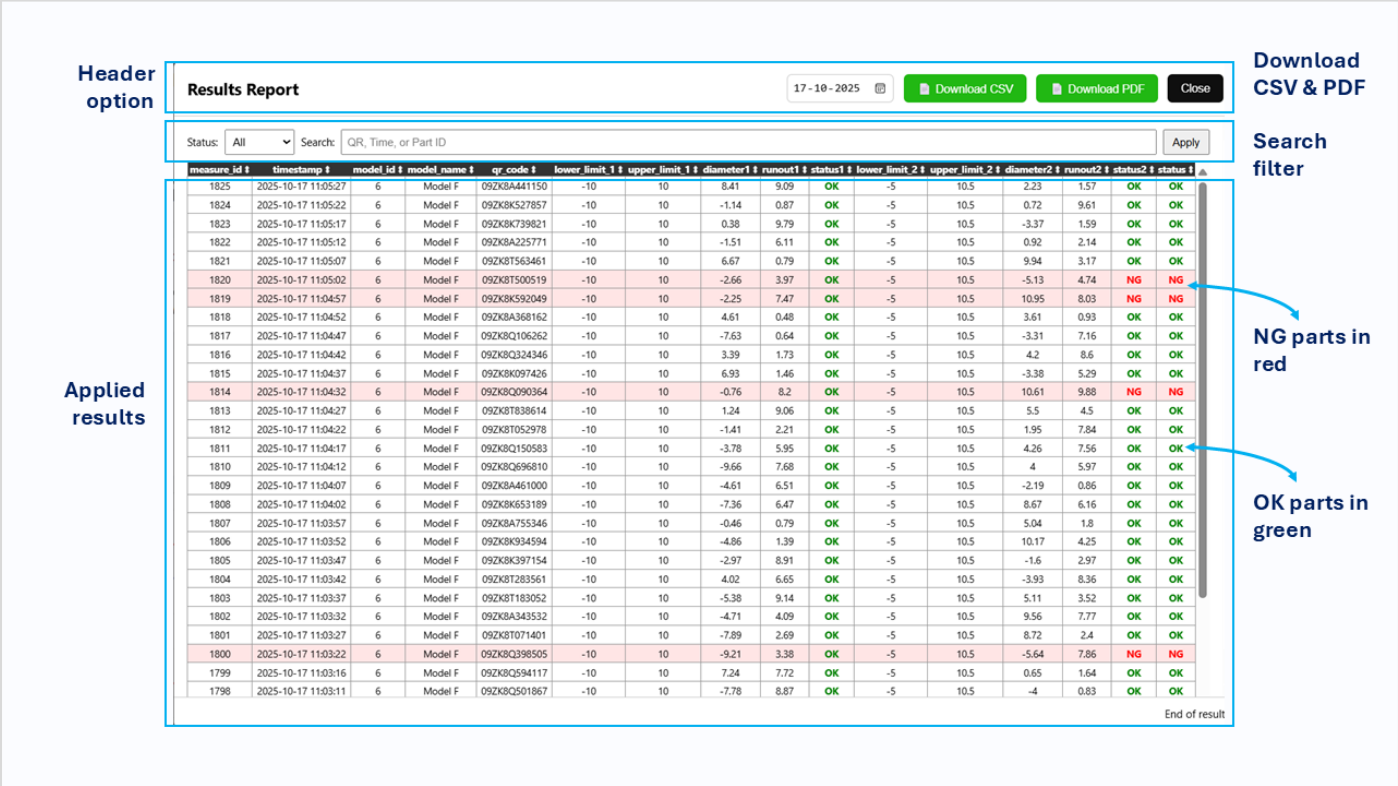
b. Model Management.

- Name, limits can be edited from here
- List all the models

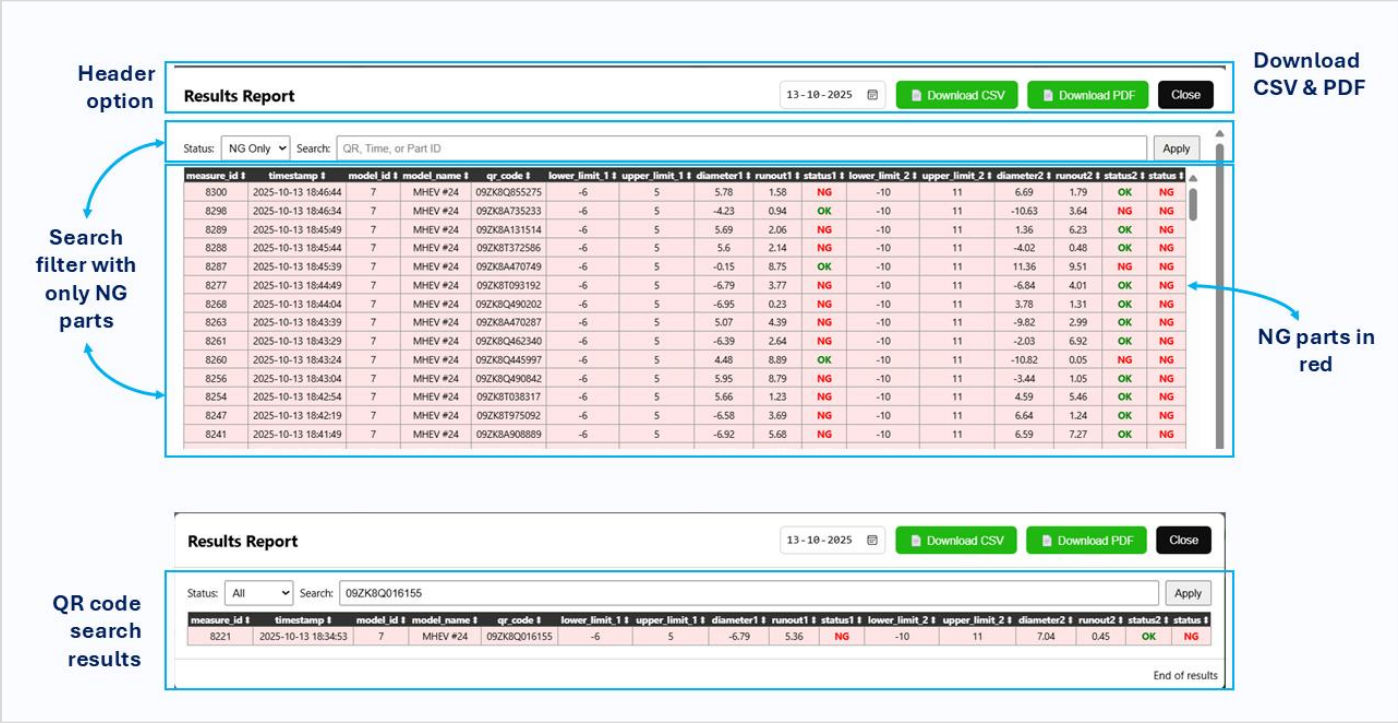


c. Result Log View

- Professional Format
- Historian data with its limits
- OK/NG highlighted in green and red



- Filters based on date, status, part ID and model name
- Can be downloaded as PDF and CSV



d. PDF Reports – Diameter results

- Result pdf format

Custom header

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IDS-HT-80-2 — Inspection Report

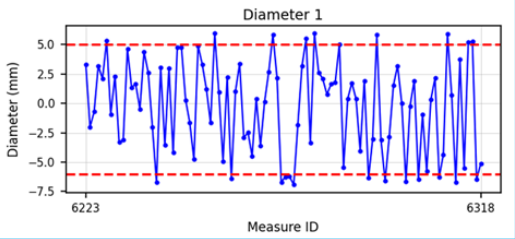
SP systems

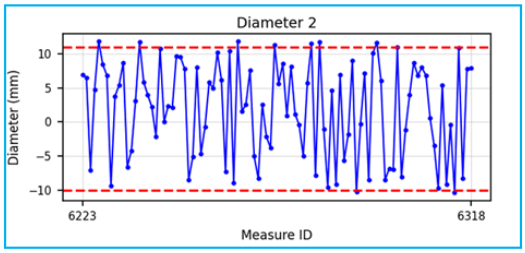
Filter options

Date: 2025-10-12 Status Filter: All Search: None

Total Parts: 96 OK: 66 NG: 30

Diameter 1 graph





Diameter 2 graph

Details of the components

measure_id	timestamp	model_id	model_name	qr_code	lower_limit_1	upper_limit_1	diameter1	runout1	status1	lower_limit_2	upper_limit_2	diameter2	runout2	status2	status
6223	2025-10-12 13:34:35	7	MHEV #24	09ZK8Q622609	-6.0	5.0	3.28	2.62	OK	-10.0	11.0	6.87	4.05	OK	OK
6224	2025-10-12 13:34:40	7	MHEV #24	09ZK8A192481	-6.0	5.0	-2.01	5.34	OK	-10.0	11.0	6.52	2.97	OK	OK
6225	2025-10-12 13:34:45	7	MHEV #24	09ZK8Q923433	-6.0	5.0	-0.7	2.28	OK	-10.0	11.0	-7.06	6.69	OK	OK
6226	2025-10-12 13:34:50	7	MHEV #24	09ZK8T517189	-6.0	5.0	3.2	7.58	OK	-10.0	11.0	4.71	3.2	OK	OK
6227	2025-10-12 13:34:55	7	MHEV #24	09ZK8T725077	-6.0	5.0	2.08	9.56	OK	-10.0	11.0	11.83	5.65	NG	NG
6228	2025-10-12 13:35:00	7	MHEV #24	09ZK8Q631479	-6.0	5.0	5.35	5.94	NG	-10.0	11.0	8.4	3.7	OK	NG
6229	2025-10-12 13:35:05	7	MHEV #24	09ZK8A392376	-6.0	5.0	-0.92	8.14	OK	-10.0	11.0	6.84	5.15	OK	OK
6230	2025-10-12 13:35:10	7	MHEV #24	09ZK8T405013	-6.0	5.0	2.32	3.88	OK	-10.0	11.0	-9.28	5.24	OK	OK
6231	2025-10-12 13:35:15	7	MHEV #24	09ZK8Q478782	-6.0	5.0	-3.29	0.21	OK	-10.0	11.0	3.8	4.91	OK	OK
6232	2025-10-12 13:35:20	7	MHEV #24	09ZK8Q711458	-6.0	5.0	-3.12	1.67	OK	-10.0	11.0	5.42	0.72	OK	OK

NG parts shaded in red

Footer

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e. PDF Reports – Diameter results

Custom header

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Capability Analysis Report

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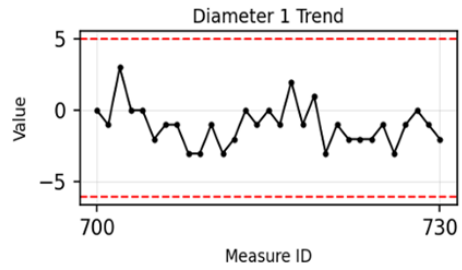
Date and filter

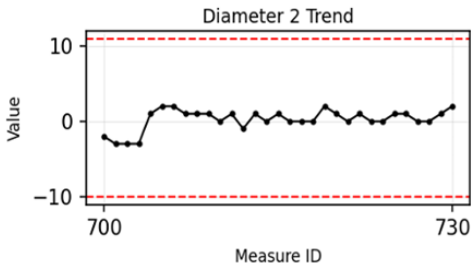
Date: 2025-10-13 21:59:10 **Include Runout:** Yes

Calculated values

Parameter	Cp	Cpk	Mean	Std. Dev
Diameter 1	0.96	0.867	-1.0323	1.9092
Diameter 2	1.859	1.816	0.2581	1.8826

Diameter trend graph for the selected values





General details about Cp/CPK

Interpretation Guide

- **Mean (μ):** Average measurement of all inspected parts.
- **Standard Deviation (σ):** Indicates the spread of the process (lower = more consistent).
- **Cp:** (USL - LSL) / (6σ) — potential capability.
- **Cpk:** min[(USL - μ)/(3σ), (μ - LSL)/(3σ)] — actual capability considering centering.